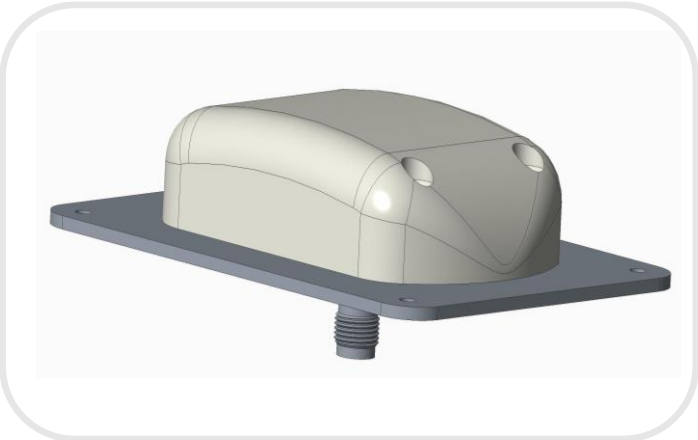
Thiagarajar Telekom Solutions (P) Limited

TRIBAND AVIONIC ANTENNA – TTSLIGGNAA-22



The TTSLIGGNAA-22 is a High Performance Rugged Triband GNSS Antenna designed to operate at GPS, GLONASS, Galileo, BeiDou and the IRNSS bands making it suitable for Multi Constellation GNSS signal reception. The Antenna has a good gain in each of the bands with a low noise figure and a high rejection at out of band frequencies making it suitable for High Precision GNSS based Navigational Applications

The antenna geometrical center and the electrical phase centers of L1, L5 and S bands are co-located in X and Y coordinates and have very minimal variations in the Z co-ordinate which helps in high precision position measurements and makes installation easier. It also includes a novel technique for suppression of multi-path signals.

The Compact design which meets the environmental conditions like marine applications the ruggedized enclosure with Aluminum Alloy (AA-2014) base and Ultem Radome. The TTSLIGGNAA-22 is housed in a pre- tapped screw-hole mount, weather-proof enclosure for permanent installations.

Key Features

- Best Signal Reception in Multi - constellation mode.
- Excellent Multipath rejection
- Good Signal to Noise ratio and Out of band Rejection
- The Low Power consumption
- Lightweight with small form factor for easier integration.

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Technical Specifications

Electrical Specifications	
IRNSS L5	1176.45 MHz (±12MHz)
IRNSS S	2492 MHz (±8.25MHz)
GPS L1	1575 MHz (±10 MHz)
GLONASS L1	1598 MHz-1606 MHz
VSWR	≤ 2.0 :1
Passive Gain	>+3.5dBiC
LNA Gain	30 ± 2 dB (L Band) 30 ± 2 dB (S Band)
LNA Power Consumption	5V - < 100mA
Out of Band Rejection	> 60dB
Noise Figure	≤ 2.0 dB
Output Impedance	50 Ohms
Axial Ratio	< 3 dB
Lightening Protection	DC Grounding
Burnout Limit	+20dBm @ CW Input
Power Handling	1 Watt
Connector	TNC - (F)

Environmental Specifications	
Mounting	Standard Screw Mounting
Finish	Polyurethane Enamel
Operating Temperature	-40°C to +85°C
Storage Temperature	-40°C to +85°C
Humidity	95% RH at 45°C
MTBF	1,00,000 Hrs.

Environmental Stress Screening	
Mechanical Vibration	MIL-STD-810G / JSS 55555
Functional Shock	MIL-STD-810G / JSS 55555
Crash hazard	MIL-STD-810G / JSS 55555
Temperature and Altitude	MIL-STD-810G / JSS 55555
Humidity	MIL-STD-810G / JSS 55555
Temperature Shock	MIL-STD-810G / JSS 55555
Salt Fog	MIL-STD-810G / JSS 55555
Fungus	MIL-STD-810G / JSS 55555
Sand and Dust	MIL-STD-810G / JSS 55555
EMI & EMC	MIL-STD-461F

Mechanical Specifications	
Dimension Excluding Connector (L x W x H)	Length 130 ±0.3mm Width 100 ±0.3mm Height 30 ±0.3 mm
Dimension Including Connector (L x W x H)	Length 130 ±0.3mm Width 100 ±0.3mm Height 46.5 ±0.3mm
Weight	295 ±10 grams.

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Note: All the Electrical Parameters are complying with Do-373 standards

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